

Assessment Brief

Module title	Software Engineering Principles
Module code	COMP2912
Assignment title	Software Design Brief - written report
Assignment type and description	Coursework
Rationale	The assignment is to develop and document a software design brief suitable to be given to a software development team so that they can establish technical feasibility, estimate effort and produce an initial quote for the work.
Page limit and guidance	The report has a limit of six pages and should be written single-spaced in an 11 pt font (or larger) with a page size of A4. Font size may be smaller in tables or diagrams provided these are still clear and legible. Note that the marker is not required to mark any more than six pages so will stop reading and marking at the page limit. You do not need to put the title, table of content, references on separate pages.
Weighting	20%
Submission deadline	16 th November 2023
Submission method	Turnitin Submission in Minerva
Feedback provision	Marking sheet with comments
Learning outcomes assessed	Select and use appropriate industry-standard tools and techniques in the process of software development
Module leader	Owen Johnson
Other Staff contact	None

1. Assignment guidance

The assignment is to develop and document a **software design brief** suitable to be given to a software development team so that they can establish technical feasibility, estimate effort and produce an initial quote for the work.

2. Assessment tasks

Scenario

A mental health charity want to support the good mental health of university students. It would like to commission an app and has set aside £100,000 for the app's development. The idea is that students concerned about their mental health could download the app and use it to record, review and get advice that will support them through the challenges of university life.

In particular the charity want to promote health eating, regular exercise and good sleep patterns as the key to good mental health. They also recognise that students can find it challenging coping with stress and think a daily diary of mood plus eating, sleeping and activities could help inform some Artificial Intelligence (AI) interventions to nudge students into healthier lifestyles. The app would be piloted at one university but should be designed for use by students at universities all over the world.

The charity have said that they want to see designs that are creative, ambitious and that make use of both integration with university systems (attendance, timetables, marks, medical records, etc) and the latest advances in smart phone technologies (location, alerts, camera, motion, embedded AI etc.). However the design must also be achievable within the budget and timescale and must embody sound software engineering principles and address any legal, ethical and sociological issues.

The charity recognise that their initial ideas are just a starting point, that they don't really know what the best business model is, or what the requirements are or even whether their ideas are feasible. They are hoping that each university will fund any necessary interfaces, promote the app to their students and make a contribution to the running costs. The ethics and privacy details are a particular concern.

The charity will pay £10,000 for the best software design and have invited you and a number of your competitors to submit six-page design briefs. They will use these to decide who will win the contract to do the full design. You will have to do the design brief for free in the hope that you get the contract to do the full design. Your design brief is therefore a very brief summary of the key features of the full design, obviously the full design is much bigger.

Your task

Investigate the range of potential requirements that you might select for your chosen solution. You have a wide range of choices available and you only need to choose some of those for your solution. Your approach is up to you but it would be sensible to do three iterations of investigation and design – the first should be a broad investigation to help you choose a candidate design, the second should investigate specific requirements to help you “flesh out” your design and the third should be “polishing” to format your design brief in the required format. Your final design brief should be in line with the client's aims, it should address a clearly articulated need, and should be technically and practically feasible.

You are expected to draw on and apply the tools, techniques and understanding you have acquired on this module. You may work with other students to discuss the range of issues and options but you must submit an individual piece of work. Each report is checked by Turnitin for plagiarism against current students, previous students and other universities as well as against online sources.

Report Structure

Please use the following report structure:

Introduction (15%)

Summarise your understanding of the project's aim and explain what you have chosen to focus on and why. Describe the investigation of requirements that you have conducted and the key points that have informed your choice of design. Who are the stakeholders and what interest do they have in the solution?

Summary of the proposed solution (15%)

Describe your proposed solution in non-technical terms. State who will use your solution, why and when. Describe what effort is required by the user to use your proposed solution and what value the user will get in return. Describe any other users and uses of the solution including any systems that you expect your solution to interface with.

Requirements (15%)

Draw a use case diagram to show the system, its users and uses. This should include any system actors. Make sure that all human users have a clear value proposal. Write short narrative that summarises the use case diagram. List any key non-functional requirements.

Using the Proposed Solution (15%)

Write a user story to illustrate how you imagine your proposed system might be used. You may choose to include some prototypes of proposed screens or an activity diagram or interaction diagram to supplement your story.

Implementing the Proposed Solution (15%)

Draw an implementation diagram (deployment and component combined) to show the hardware and network infrastructure for your proposed solution together with the software components that will be required. Describe what will be required in terms of both new and existing hardware, networking, software and interfaces to implement the system. What technology platforms will be used? How will the non-functional requirements be addressed?

Conclusion (15%)

Identify any legal, ethical and sociological issues and explain how your solution will address them. Make a conclusion explaining why your proposal follows sound software engineering principles and is achievable within the budget and timescale. What are the limitations of your proposed solution and what outstanding issues have not been addressed?

Report Writing and Presentation (10%)

Your report should be logically structured, use good English, use neat diagrams where appropriate, include accurate and consistent referencing where appropriate, use correct spelling and punctuation.

3. General guidance and study support

Please see the Module Handbook available in Minerva for general guidance and support. A range of guidance was provided in COMP1911 for technical report writing. Although this is a different type of document much of the guidance for good technical report writing still applies. For example you should still give your report a title and make sure to reference facts and figures from external information sources using Leeds Harvard or Leeds Numeric referencing style.

4. Assessment criteria and marking process

Your assignment should demonstrate that you can:

- Apply a range of tools and approaches from the module to design a system.
- Understand how to elicit and develop requirements and design novel and appropriate solutions.
- Structure a report based on your design.
- Demonstrate an ability to work as a software engineer.

Your report should:

- Be logically structured.
- Use good English.
- Use neat diagrams where appropriate.
- Include accurate and consistent referencing where appropriate.
- Use correct spelling and punctuation.

5. University presentation and referencing

The quality of written English will be assessed in this work following the University Pass/ Fail criteria. As a minimum, you must ensure:

- Paragraphs are used.
- There are links between and within paragraphs although these may be ineffective at times.
- There are (at least) attempts at referencing.
- Word choice and grammar do not seriously undermine the meaning and comprehensibility of the argument.
- Word choice and grammar are generally appropriate to an academic text.

These are pass/ fail criteria. So irrespective of marks awarded elsewhere, if you do not meet these criteria you will fail overall.

6. Submission requirements

You must submit your assignment as a Word document or pdf via the Turnitin link in the Assessment folder for the module in Minerva. Late submissions without mitigating circumstance will incur a 5% penalty per day.

7. Academic misconduct and plagiarism

Leeds students are part of an academic community that shares ideas and develops new ones. You need to learn how to work with others, how to interpret and present other people's ideas, and how to produce your own independent academic work. It is essential that you can distinguish between other people's work and your own, and correctly acknowledge other people's work. All students new to the University are expected to complete an online [Academic Integrity tutorial and test](#), and all Leeds students should ensure that they are aware of the principles of Academic integrity.

When you submit work for assessment it is expected that it will meet the University's academic integrity standards.

If you do not understand what these standards are, or how they apply to your work, then please ask the module teaching staff for further guidance.

By submitting this assignment you are confirming that the work is a true expression of your own work and ideas and that you have given credit to others where their work has contributed to yours.

8. Assessment/ marking criteria grid

Assignments will be marked using the following marking sheet.

	Excellent	Good	Pass	Fail
	3	2	1	0
Introduction <i>Summarise your project's aim and explain what you have chosen to focus on and why.</i> <i>Describe the investigation of requirements that you have conducted (interviews, online research, observation etc) and the key points that have informed your choice of design.</i> <i>Who are the stakeholders and what interest do they have in the solution?</i>				
Summary of the proposed solution <i>Describe your proposed solution in non-technical terms.</i> <i>State who will use your solution, why and when.</i> <i>Describe what effort is required by the user to use your proposed solution and what value the user will get in return.</i> <i>Describe any other users and uses of the solution including any systems that you expect your solution to interface with.</i>				
Requirements <i>Draw a use case diagram to show the system, its users and uses. This should include any system actors.</i> <i>Make sure that all human users have a clear value proposal.</i> <i>Write a short narrative that summarises the use case diagram.</i> <i>List any key non-functional requirements.</i>				
Using the Proposed Solution <i>Write a user story to illustrate how you imagine your proposed system might be used.</i> <i>You should include</i> <i> some prototypes of proposed screens</i> <i> or an activity diagram</i> <i> or interaction diagram to supplement your story.</i>				
Implementing the Proposed Solution <i>Draw an implementation diagram (deployment and component combined) to show the hardware and network infrastructure for your proposed solution together with the software components that will be required.</i> <i>Describe what will be required in terms of both new and existing hardware, networking, software and interfaces to implement the system.</i> <i>What technology platforms will be used?</i> <i>How will the non-functional requirements be addressed?</i>				
Conclusion <i>Identify the legal, ethical and sociological issues and explain how your solution will address them.</i> <i>Make a conclusion explaining why your proposal follows sound software engineering principles and is achievable within the budget and timescale. What are the limitations of your proposed solution and what outstanding issues have not been addressed?</i>				
Report Writing and Presentation <i>Your report should be logically structured, use good English, use neat diagrams where appropriate, include accurate and consistent referencing where appropriate, use correct spelling and punctuation.</i>				
University minimum standards for presentation and referencing achieved (Pass/ Fail)	(Pass/ Fail)			
Total marks (out of 20 marks available)				

Markers apply the following rubric when determining the mark for each section:

Excellent	3	Excellent, concise and comprehensive content with strong evidence of novelty and originality. Systematic and thorough for all components with exemplary use of relevant methods.
Good	2	Good, clear statements and explanations with some evidence of novelty and originality. Most components covered to a high standard with largely correct use of relevant methods.
Pass	1	Weak but still effective explanations but with limited novelty and originality. Most components covered with an acceptable use of relevant methods.
Fail	0	Poor or confusing content, little evidence of novelty or originality. Major components missing or covered with major errors in relevant methods.